DSA REFERENCE

Backtracking

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Build one candidate incrementally, abandon invalid prefixes early, and undo each choice before trying the next.

VERSION 1.0 REVIEW
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STATUS HUMAN REVIEW REQUIRED

DSA / BACKTRACKING

DSA

Backtracking

CORE IDEA

Build one candidate incrementally, abandon invalid prefixes early, and undo each choice before trying the next.

Recognition cues

+ enumerate combinations or permutations
+ constraint satisfaction
+ all valid arrangements
+ small exponential search

Implementation sequence

+ define choice set
+ prune before descending
+ choose
+ recurse
+ undo symmetrically

Invariant

The current path is a valid partial candidate and all mutable state reflects exactly that path.

Worked reasoning

For combinations summing to a target, sort candidates, skip duplicates at the same depth, and stop when the remainder is negative.

Complexity

Output-sensitive exponential time; stack and path space are bounded by decision depth.

Failure modes

- [ ] forgetting to undo
- [ ] copying excessive state
- [ ] duplicate outputs
- [ ] weak pruning
- [ ] confusing subsequences with permutations

Practice ladder

Backtracking: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Subsets**	State the cue, invariant, edge case, and proof before implementation.
2	**Permutations**	State the cue, invariant, edge case, and proof before implementation.
3	**Combination Sum**	State the cue, invariant, edge case, and proof before implementation.
4	**Word Search**	State the cue, invariant, edge case, and proof before implementation.
5	**N-Queens**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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